A Five-Cycle Obstruction for the Positive Lyness Map

Route-A structural certificate C173

Abstract

For the positive-quadrant map $F(x, y) = (y, (1 + y)/x)$, direct rational simplification gives $F^5 = I$. There is one fixed point, (ϕ, ϕ) with $\phi = (1 + \sqrt{5})/2$, and every other point has exact period five. Hence $\mathrm{Fix}(F^n)$ is a singleton when $5 \nmid n$ and the entire positive quadrant when $5 \mid n$. Thus the classical Artin–Mazur series is undefined. The measure $dx\,dy/(xy)$ is invariant, and the resulting Koopman operator is a natural unitary of order five. It is not trace class, so the ordinary Fredholm determinant is unavailable as well. These exact obstructions reject the model as a periodic-orbit zeta candidate; no regularized substitute or target-spectrum claim is made.

Keywords: Lyness map; birational dynamics; exact periodicity; fixed-point obstruction; Artin–Mazur zeta; Route A.

中文摘要

正象限映射 $F(x, y) = (y, (1+y)/x)$ 经有理函数直接化简满足 $F^5 = I$。 其唯一不动点为 (ϕ, ϕ)，其中 $\phi = (1+\sqrt{5})/2$；其余每一点 都具有严格周期五。因此，当 $5 \nmid n$ 时，$\mathrm{Fix}(F^n)$ 只含一个点；当 $5 \mid n$ 时，它等于整个正象限。经典 Artin–Mazur 级数因而没有定义：第五项需要 一个不可数不动点集的基数。测度 $dx\,dy/(xy)$ 保持不变，由此得到的 Koopman 算子是五阶自然酉算子，但并非迹类， 所以普通 Fredholm 行列式同样不可用。本文不引入正则化替代物，也不提出目标谱结论。

关键词：Lyness 映射；双有理动力学； 精确周期性；不动点障碍；Artin–Mazur zeta；Route A。

1 Exact five-cycle identity

On $X = (0, \infty)^2$, every denominator below is nonzero. Direct substitution gives

$$F(x, y) = \left(y, \frac{1 + y}{x}\right), \qquad F^2(x, y) = \left(\frac{1 + y}{x}, \frac{1 + x + y}{xy}\right),$$

$$F^3(x, y) = \left(\frac{1 + x + y}{xy}, \frac{1 + x}{y}\right), \qquad F^4(x, y) = \left(\frac{1 + x}{y}, x\right), \qquad F^5(x, y) = (x, y).$$

Thus $F^5 = I$ globally, not merely on a finite sample.

Proposition. The point (ϕ, ϕ) is the unique fixed point, and every other point of X has exact least period five.

Proof. A fixed point has $y = x$ and $x^2 = x + 1$. Only the root $\phi = (1 + \sqrt{5})/2$ is positive. Every least period divides the prime number five, so deleting the unique period-one point leaves period five. □

2 The fixed-set obstruction

For every positive integer n,

$$\mathrm{Fix}(F^n) = \begin{cases} \{(\phi, \phi)\}, & 5 \nmid n, \\ X, & 5 \mid n. \end{cases}$$

The classical definition

$$\zeta_{\mathrm{AM}}(z) = \exp\left(\sum_{n \geq 1} \#\,\mathrm{Fix}(F^n) \frac{z^n}{n}\right)$$

requires every fixed set to be finite. Here $\mathrm{Fix}(F^5) = X$ is uncountable. Replacing it by a component count, a quotient count, or a regularized trace would define a different object and is not done here.

3 Invariant geometry and natural lift

The Jacobian determinant is $(1 + y)/x^2$, while the product of the image coordinates is $y(1 + y)/x$. Therefore

$$\frac{|\det DF(x, y)|}{F_1(x, y)F_2(x, y)} = \frac{1}{xy},$$

so $d\mu = dx\,dy/(xy)$ is invariant. Moreover

$$F^{-1}(x, y) = \left(\frac{1 + x}{y}, x\right), \qquad R(x, y) = (y, x), \qquad RFR = F^{-1}.$$

On $H = L^2(X,\mu)$ define $Uf = f \circ F$. Then U is unitary and $U^5 = I$. If $\omega = e^{2\pi i/5}$, cyclic Fourier orthogonality gives

$$P_j = \frac{1}{5}\sum_{r=0}^{4} \omega^{-jr} U^r, \qquad \operatorname{ran} P_j = \ker(U - \omega^j I).$$

The negative sign is fixed by $UP_j = \omega^j P_j$.

The nonatomic free action off (ϕ, ϕ) supplies infinitely many disjoint positive-measure orbit tubes. Localizing a nonzero function on each tube and applying any P_j gives mutually orthogonal nonzero vectors. Thus every fifth-root eigenspace is infinite-dimensional. Hence U is noncompact, belongs to no finite Schatten class, and is not trace class; the ordinary trace-class determinant $\det(I - zU)$ does not exist. It is also not self-adjoint: self-adjoint unitarity and $U^5 = I$ would force $U = I$, contrary to the nontrivial composition action.

4 Boundary

The finite rational ledger in the release is a regression sentinel, not a proof. No prime correspondence, target divisor, functional equation, counting law, arithmetic local datum, Euler factor, root number, automorphy, Hilbert–Pólya operator, or Route-B authorization is claimed. The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations. All data and code are included. No external dataset, human or animal subject, or private data is involved. The anonymous contribution covers conceptualization, proof, software, validation, and writing. No funding or competing interest is declared. AI assistance was used for drafting and code generation; all claim-bearing formulas were independently reconstructed, and no external reviewer or acceptance score is represented.